

Basics 21: Single-Fluid and Multifluid Descriptions

III - Elementary plasma physics

Species equations, collective variables, generalized Ohm's law, and the hierarchy of plasma-fluid models

Textbook draft, version 1.0. A plasma contains multiple species, yet many macroscopic phenomena are described by one bulk fluid. This chapter shows how species continuity and momentum equations are combined into center-of-mass, charge, and current variables, and it identifies the approximations separating multifluid, Hall-MHD, and ideal-MHD models. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Write species continuity and momentum equations.
- Define mass density, bulk velocity, charge density, and current.
- Derive the summed continuity and momentum equations.
- Obtain the structure of generalized Ohm's law from electron momentum balance.
- Distinguish single-fluid, two-fluid, Hall-MHD, and ideal-MHD regimes.
- Recognize the need for pressure and heat-flux closures.

2 Prerequisites and reading strategy

- Basics 3, 5, 13, 17, and 19.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Species fluid equations

For each species s with mass m_s , charge q_s , density n_s , velocity $\mathbf{u}_s$, and pressure tensor $\mathbf{P}_s$,

$$\frac{\partial n_s}{\partial t} + \nabla \cdot (n_s \mathbf{u}_s) = S_s, \quad (3.1)$$

$$m_s n_s \left(\frac{\partial \mathbf{u}_s}{\partial t} + \mathbf{u}_s \cdot \nabla \mathbf{u}_s \right) = q_s n_s (\mathbf{E} + \mathbf{u}_s \times \mathbf{B}) - \nabla \cdot \mathbf{P}_s + \mathbf{R}_s + \mathbf{M}_s. \quad (3.2)$$

$\mathbf{R}_s$ represents collisional momentum exchange and $\mathbf{M}_s$ other sources. These equations are exact moment equations only when higher moments are carried consistently.

4 Collective variables

Define

$$\rho = \sum_s m_s n_s, \quad (4.1)$$

$$\rho_q = \sum_s q_s n_s, \quad (4.2)$$

$$\rho \mathbf{u} = \sum_s m_s n_s \mathbf{u}_s, \quad (4.3)$$

$$\mathbf{J} = \sum_s q_s n_s \mathbf{u}_s. \quad (4.4)$$

The bulk velocity is mass weighted, whereas current is charge weighted. In an electron-ion plasma with $m_e \ll m_i$, $\mathbf{u}$ lies close to the ion velocity even when electrons carry most of the current.

5 Single-fluid mass conservation

Multiply each species continuity equation by m_s and sum:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = \sum_s m_s S_s. \quad (5.1)$$

For ionization and recombination internal to the modeled plasma, total mass is conserved even though individual species populations change, so the right-hand side can vanish.

6 Summed momentum equation

Summing species momentum equations cancels internal collisional exchange:

$$\sum_s \mathbf{R}_s = 0. \quad (6.1)$$

The electromagnetic force becomes

$$\sum_s q_s n_s (\mathbf{E} + \mathbf{u}_s \times \mathbf{B}) = \rho_q \mathbf{E} + \mathbf{J} \times \mathbf{B}. \quad (6.2)$$

Under quasineutrality the electric-force term is small, giving schematically

$$\rho \frac{D\mathbf{u}}{Dt} = \mathbf{J} \times \mathbf{B} - \nabla \cdot \mathbf{P} + \text{relative-drift corrections}. \quad (6.3)$$

A strict derivation retains the stress associated with species velocities relative to $\mathbf{u}$.

7 Electron momentum and generalized Ohm's law

For singly charged electrons,

$$m_e n_e \frac{D_e \mathbf{u}_e}{Dt} = -en_e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B}) - \nabla \cdot \mathbf{P}_e + \mathbf{R}_e. \quad (7.1)$$

Solving for $\mathbf{E}$ and writing approximately $\mathbf{u}_e = \mathbf{u} - \mathbf{J}/(en_e)$ yields

$$\mathbf{E} + \mathbf{u} \times \mathbf{B} = \frac{\mathbf{J} \times \mathbf{B}}{en_e} - \frac{\nabla \cdot \mathbf{P}_e}{en_e} + \frac{\mathbf{R}_e}{en_e} - \frac{m_e}{e} \frac{D_e \mathbf{u}_e}{Dt}. \quad (7.2)$$

The terms are Hall, electron-pressure, resistive/collisional, and electron-inertia contributions. Their ordering determines the fluid model.

8 Model hierarchy

Multifluid: evolves each species velocity and thermodynamics.

Two-fluid: commonly retains distinct ion and electron dynamics.

Hall MHD: keeps $\mathbf{J} \times \mathbf{B}/(en_e)$ in Ohm's law.

Resistive MHD: keeps an effective $\eta \mathbf{J}$ term.

Ideal MHD: assumes $\mathbf{E} + \mathbf{u} \times \mathbf{B} = 0$.

These are ordered reductions, not interchangeable software options. A model is selected by comparing frequencies, length scales, collision rates, and gyro scales.

9 Closure and loss of kinetic information

Fluid equations contain pressure tensors and heat fluxes whose evolution introduces still higher moments. A scalar equation of state,

$$p = p(\rho, S), \quad (9.1)$$

or an adiabatic law closes ideal MHD, but it removes resonant velocity-space physics, finite-Larmor-radius effects, and non-Maxwellian energetic-particle drive. Kinetic MHD restores selected kinetic responses while retaining a fluid bulk plasma.

10 Computational laboratory

Given electron and ion densities and velocities, compute ρ , ρ_q , $\mathbf{u}$, and $\mathbf{J}$, then reconstruct species velocities under quasineutrality. Build an ordering calculator comparing Hall, electron-inertia, pressure, and resistive terms for several characteristic scales.

11 Summary

- Multifluid equations preserve species-specific motion; single-fluid variables reorganize them into bulk and current dynamics.
- Mass-weighted flow and charge-weighted current contain different information.
- Generalized Ohm's law is the central discriminator among fluid plasma models.
- Ideal MHD follows only after Hall, pressure, inertia, and resistive terms are ordered small.
- Closure assumptions determine which kinetic effects the fluid model cannot represent.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
ρ	mass density	$\sum_s m_s n_s$.
ρ_q	charge density	$\sum_s q_s n_s$.
$\mathbf{u}$	bulk velocity	Mass-weighted species velocity.
$\mathbf{J}$	current density	Charge-weighted species flux.
$\mathbf{P}_s$	species pressure tensor	Second central velocity moment.
$\mathbf{R}_s$	collisional momentum exchange	Internal interspecies force density.

12 Exercises

1. Derive total mass conservation by summing species continuity equations.
2. Show how the species Lorentz forces combine into $\rho_q \mathbf{E} + \mathbf{J} \times \mathbf{B}$.
3. Derive the Hall term using $\mathbf{u}_e \simeq \mathbf{u} - \mathbf{J}/(en_e)$.

4. Explain why bulk velocity is generally closer to ion velocity than electron velocity.
5. Identify physics lost when a pressure tensor is replaced by scalar pressure.

13 References

References

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